GENERICS

```
// class
class Box<T> { }
// variable
Box<String> b = new Box<String>();
// method
public <T> void doStuff(T value) { }
// Full Example
class OrderedPair<T, U> {
  T first;
  U second;
  public OrderedPair(T t, U u) {
    this.first = f;
    this.second = s;
  }
  public T getFirst() { return first; }
  public U getSecond() { return second; }
  @Override
  public int hashCode() {
      return Objects.hash(first.hashCode(),
                          second.hashCode());
  }
  @Override
  public boolean equals(Object obj) {
      if (obj == null ||
          obj.getClass() != getClass()) return false;
      @SuppressWarnings("unchecked")
      var other = (OrderedPair<T, U>)obj;
      return Objects.equals(first, other.first) &&
             Objects.equals(second, other.second);
  }
}

public class Employee implements Comparable<Employee>
{
  // ...
  @Override
  public int compareTo(Employee other) {
    return Double.compare(this.salary, other.salary);
  }
}
```

JAVA BEING STUPID

```
T obj = new T();            // error
obj instanceof T;           // error
T heh = (T) "asdf";         // no typechecking
var s = Stack<int>();       // no primitive types
private static T x;         // static no workey
Node<Number> s = new Node<Integer>(); // error
class IThrowAnErrorBecauseWhyNot {
  void m(List<String> list) { }
  void m(List<Integer> list) { }
} // error because of type erasure & identifiability
static <T extends Comparable<T>> T m(T x, T y, T z) {
  return x > y ? z : x;
}
Integer n = m(1, 3.141, 0);          // error
Number n = m(1, 3.141, 0);           // ok
```

COMPARABLE

```
interface Comparable<T> {
  int compareTo(T o);
}
static <T extends Comparable> T doStuff(T a, T b) {
  // compareTo is possible with all types
  return a.compareTo("b") > 0 ? a : b;
}
static <T extends Comparable<T>> T doStuff(T a,T b) {
  // compareTo is possible only with T
  return a.compareTo(b) > 0 ? a : b;
}
```

EXAMPLE

```
class Graphic {
  public void draw() { }
}
class Rectangle extends Graphic { }
class Circle extends Graphic { }
class GraphicStack<T extends Graphic>
    extends Stack<T> {
  public void drawAll() {
    for (T item : this) item.draw();
  }
}
```

BYTECODE

JVM Architecture:

– Operand Stack (LIFO op vals)
– Local variables (op results)
– Constant pool (runtime refs)

TYPE ERASURE

– Introduced because of "bAcKwArDs CoMpAtAbIlItY"
– No type information at runtime (Non-Reifiable Type)

```
class MyStack<T> {
  Entry<T> top;
  int push(T value) { }
}
// becomes
class MyStack {
  Entry top;
  int push(Object value) { }
}
/* with bounds */
class MyStack<T extends Comparable<T>> {
  Entry<T> top;
  int push(T value) { }
}
// becomes
class MyStack {
  Entry top;
  int push(Comparable value) { }
}
/* what happens at/before runtime */
MyStack<String> s = new MyStack<String>();
s.push("Eh");
String x = s.pop();
// becomes
MyStack s = new MyStack();
s.push("Eh");
String x = (String) s.pop();
```

WILDCARDS

```
void printGs(List<Graphic> gs) { }
List<Graphic> gs1 = new ArrayList<>();
printGs(gs1);   // ok
List<Circle> gs2 = new ArrayList<>();
printGs(gs2);   // error
/* solution */
void printGs(List<? extends Graphic> gs) { }
```

GENERIC VARIANCE

	Type	*Compatible Type-Args*	*R*	*W*
Invariance	C<T>	T	✓	✓
Covariance	C<? extends T>	T and sub-types	✓	✗
Contravariance	C<? super T>	T and basis-types	✗	✓
Bivariance	C<?>	All	✗	✗

INVARIANCE

```
static <T> void move(Stack<T>, from, Stack<T> to) {
  while(!from.isEmpty()) to.push(from.pop());
}
var rectS = new Stack<Rectangle>();
var graphS = new Stack<Graphic>();
graphS.push(new Rectangle());            // ok
move(rectS, graphS);                     // error
Stack<Object> objS = graphS;             // error
/* anotherone */
String[] strA = new String[10];
Object[] objA = strA;                    // ok
objA[0] = Integer.valueOf(2);            // runtime err
/* anotherone */
ArrayList<String> sA = new ArrayList<>();
ArrayList<Object> oA = sA;               // error
// error
List<Graphic> l = new ArrayList<Rectangle>();
// ok
Rectangle[] rectangleStack = new Rectangle[10];
Graphic[] graphicStack = new Graphic[10];
graphicStack = rectangleStack;
// error
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
graphicStack = rectangleStack;
// ok
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
for (Rectangle r : rectangleStack)
  graphicStack.push(r);
```

COVARIANCE

```
public class Stack<E> {
  public void pushAll(Iterable<? extends E> src) { }
}
Stack<? extends Graphic> graphS;
graphS = new Stack<Rectangle>();    // ok
graphS = new Stack<Circle>();       // ok
graphS = new Stack<Object>();       // error (duh)
/* read only */
graphS.push(new Graphic());         // error
graphS.push(new Rectangle());       // error
graphS.push(new Triangle());        // error
```

CONTRAVARIANCE

```
static <T> void addToC(List<? super T> list, T e) {
  list.add(e);
}
addToC(new ArrayList<Number>(), 3); // ok
addToC(new ArrayList<Object>(), 3); // ok
/* shenanigans */
Stack<? super Graphic> s = new Stack<Object>();
s.add(new Graphic());               // ok
s.add(new Rectangle());             // ok
Graphic g = s.pop();                // error
Object g = s.pop();                 // ok
```

BIVARIANCE

– If concrete type doesn't matter

```
static void printList(List<?> list) {
  for (Object elem : list)
    System.out.print(elem + " ");
}
/* more shenanigans */
static void appendNewObject(List<?> list) {
  list.add(new Object());           // error
}
```

ANOTHERONE

```
public class VarianceExamples {
  public static double sum(
      Collection<? extends Number> numbers) {
    double sum = 0;
    for (Number num : numbers)
      sum += num.doubleValue();
    return sum;
  }
  public static void addNumbers(List<? super Integer>
      list, Collection<? extends Integer> source) {
    list.addAll(source);
  }
  public static <T extends Comparable<T>> T
      findMax(Collection<T> coll) {
    return coll.stream()
             .max(Comparator.naturalOrder())
             .orElse(null);
  }
  public static <T> List<T> filterByType(
      Collection<?> source, Class<T> clazz) {
    List<T> result = new ArrayList<>();
    for (Object obj : source)
      if (clazz.isInstance(obj))
        result.add(clazz.cast(obj));
    return result;
  }
}
```

TYPICAL "DOES THIS COMPILE MR. LINTER?" QUESTIONS

```
public class GenericsSyntax<T extends Iterable<T>> {
    public void fun(Iterable<? super T> i) { } // OK
}
// OK
List<? extends Object> v2 = new ArrayList<String>();
```

PRODUCER / CONSUMER

from **produces** entries, to **consumes** entries

```
<T> void move(Stack<? extends T> from,
              Stack<? super T> to) {
  while (!from.isEmpty())
    to.push(from.pop());
}
```

MISC

```
<T extends Comparable & Collection> // multiple type bounds

Set<Integer> set = new Set<>();
Integer[] c = set.toArray(new Integer[0]);
```

RECORDS

```
public record Person (String name, String address) {}
```

GENERICS STREAM TOMFOOLERY

```
List<? extends Media> mediaList;
public <S extends T> List<S>
  filterBySubtype(Class<S> subtype) {
    return mediaList.stream()
            .filter(subtype::isInstance)
            .map(subtype::cast)
            .collect(Collectors.toList());
}
```

ANNOTATIONS

– Metadata
– Macros for the poor people forced to work corporate
– @Override, @Test, @Json...

DEFINING

```
@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface Author {
  String name();
  String date() default "N/A";
}

@Author(name = "UwU", date = "idon'tvalidateinput")
public class WeeOoo { }
```

TODO:
Examples Woche03

REFLECTION

– Information of Class (private and public): Methods, Attributes, Parent class, Implemented interfaces
– Calling methods possible
– Usages: Debugger, Serialization, Frameworks, Remote Procedure Call, ORM

```
// all of these `throws SecurityException`
public Field[] getDeclaredFields()
public Method[] getDeclaredMethods()
public Constructor<?>[] getDeclaredConstructors()

System.out.println(dump(new Student("a", "b"), 0));
// {
//   firstName = a
//   lastName = b
// }
Class c = "s".getClass(); // java.lang.String
```

METHOD

```
@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.METHOD)
public @interface Init { }
// ...
public class User {
  private String needsInit;
  @Init
  private void initObj() {
    this.needsInit = "wowie";
  }
}
```

CLASS

TODO:
getDeclaredConstructor().newInstance()

```
@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.TYPE)
public @interface JsonSerializable { }
// ...
@JsonSerializable
public class User { }
```

ATTRIBUTE

```
@Retention(RetentionPolicy.RUNTIME)
@Target({ ElementType.FIELD })
public @interface JsonElement {
    public String key() default "";
}
// ...
public class User {
    @JsonElement
    private String firstName;
}
```

VALIDATION

```
@Min(value = 18, message = "Age must be ≥ 18")
@Max(value = 99, message = "Age must be ≤ 99")
private int age;
@NotNull(message = "Name cannot be null")
private String name;
```

EXAMPLE

```
@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface WebController { }
// ...
@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.METHOD)
public @interface Get {
    String path();
    String contentType() default "application/json";
}
// ...
@WebController
public class Controller {
  private User user = new User("frank", "meier");
  @Get(path = "/user")
  public String user() {
    return new ObjectToJsonConverter()
               .convertToJson(user);
  }
}
// ...
public class WebFramework {
  public void addController(Class<?> controllerClass)
              throws Exception {
    if (!controllerClass
          .isAnnotationPresent(WebController.class))
      throw new Exception("Is not a WebController");

    for (Method method : controllerClass
                   .getDeclaredMethods())
      if (method.isAnnotationPresent(Get.class)) {
        method.setAccessible(true); // best practice or
sth, idk, i write code in actually good languages
        var annot = method.getAnnotation(Get.class);
        routeHandlers.put(
          annot.path(),
          new WebHandler<>(method,
                   controllerClass
                     .getDeclaredConstructor()
                     .newInstance(),
                   annot.contentType())
        );
      }
  }
}
```

ALGORITHMS

Brute-force: slow, stupid, go shame yourself
Greedy: Take best option each step (optimize locally). Fast. Not necessarily global optimum
Backtracking: Trial and error. Best overall option. Higher compute costs
Divide and conquer: Binary search
Dynamic programming: Recursion optimization? Tabulation. Iterative reuse of previous results

BACKTRACKING

TODO:
slides 87

KNIGHTSTOUR

1	6	15	10	21
14	9	20	5	16
19	2	7	14	11
8	13	24	17	4
H	18	3	12	23

```java
boolean tour(int[][] visited, int x,int y, int pos) {
  visited[x][y] = pos;
  if (pos >= N * N) return true;
  for (int k = 0; k < 8; k++) {
    int newX = x + row[k];
    int newY = y + col[k];
    if (isValid(newX, newY)
        && visited[newX][newY] == 0
        && tour(visited, newX, newY, pos + 1))
        return true;
  }
  visited[x][y] = 0;
  return false;
}
```

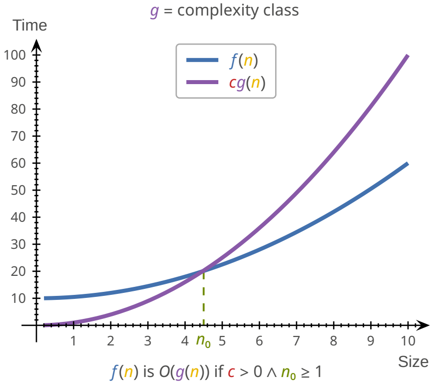
BIG O NOTATION

- Upper bound of f
- Worst case scenario
- Atomic operations = constant time
- Runtime measured as sum of primitive operations

Notation

f	Algorithm	n	Size of Problem
g	Complexity Class	c	> 0
n_0	≥ 1		

$f(n)$ = Number of steps
g = complexity class



$f(n)$ is $O(g(n))$ if $c > 0 \land n_0 \geq 1$
$f(n) \leq cg(n)$ for $n \geq n_0$

f in order g:

$$f, g : \mathbb{N} \to \mathbb{N}$$
$$n_0, c \in \mathbb{N} \land \forall n \geq n_0.\ (f(n) \leq c \cdot g(n))$$
$$\Rightarrow f \in O(g) \Leftrightarrow f = O(g)$$
$O(n)$ = Complexity class

RULES

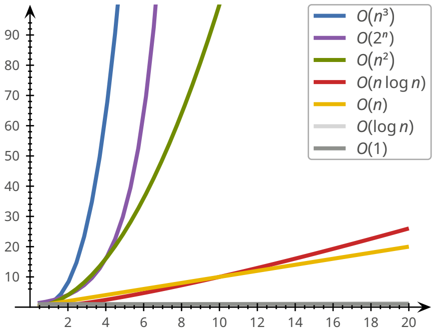
If $f(n)$ is polynomial of degree d, then $f(n) \in O(n^d)$
- ignore lower powers
- ignore constants

Use most optimal function (lowest power)
- $2n$ is $O(n)$ better than $2n$ is $O(n^2)$

Simplify as far as possible
- $3n+5$ is $O(n)$ instead of $3n+5$ is $O(3n)$

COMPLEXITY CLASSES

Name	*Class*	*Example*
Constant	1	Array read
Logarithmic	$\log n$	Binary search
Linear	n	Linear search
Log-linear	$n \log n$	Merge+Quick sort
Quadratic	n^2	Bubble+Insertion sort
Cubic	n^3	Solve equation
Polynomial	n^k	Various algorithms
Exponential	2^n	Calculate all subsets
Factorial	$n!$	Calculate all permutations



MAFS

$$1+2+\ldots+n = \sum_{k=1}^{n} = \frac{n(n+1)}{2}$$
$$1+q+q^2+\ldots+q^n = \sum_{k=0}^{n} q^k = \frac{1-q^{n+1}}{1-q}$$
$$\overset{q=2, n=\log_2(n)}{=} 2n-1$$

SORTING ALGORITHMS

BOGOSORT

The best sorting algorithm

STALINSORT

The most efficient sorting algorithm

SELECTION SORT

- Search for smallest elem in unsorted part and move in front
- Less mutations in array
- Same performance even if list almost sorted

1	3	9	6	2
1	2	3	9	6
1	2	3	9	6
1	2	3	6	9

```java
public static void selectionSort(int[] a) {
  int n = a.length;
  for (int i = 0; i < n - 1; i++) {
    int minimum = i;
    for (int j = i + 1; j < n; j++)
      if (a[j] < a[minimum])
        minimum = j;
    swap(a, i, minimum);
  }
}
```

$$(n-1)+(n-2)+\ldots+1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

INSERTION SORT

- Take elem and put into correct place in sorted part
- Good for already partially sorted arrays

1	3	9	6	2
1	3	9	6	2
1	3	9	6	2
1	3	6	9	2

```java
public static void insertionSort(int[] a) {
  int n = a.length;
  for (int i = 1; i < n; i++)
    for (int j = i; j > 0 && a[j] < a[j - 1]; j--)
      swap(a, j, j - 1);
}
```

$$1+\ldots+(n-1)+n = \sum_{i=1}^{n} i = \frac{n(n+1)}{2} = \frac{n^2+n}{2} \sim O(n^2)$$

BUBBLE SORT

- Compare neighboring elems and swap if needed

1	3	9	6	2
1	3	9	6	2
1	3	9	6	2
1	3	6	9	2

```java
public static void bubbleSort(int[] a) {
  for (n = a.length; n > 1; n--)
    for (i = 0; i < n - 1; i++)
      if (a[i] > a[i + 1])
        swap(a, i, i + 1);
}
```

$$(n-1)+(n-2)+\ldots+1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

COUNTING SORT

- For int-like data only i guess
- Count how often (n) values of index i appear in array, then insert i n times in new array

1	3	2	0	2
1	1	2	1	0
0	1	2	2	3

```java
public static void countingSort(int[] a) {
  int k = Arrays.stream(arr).max().getAsInt();
  int[] c = new int[k + 1];
  for (int i : a) c[i] += 1;
  int pos = 0;
  for (int i = 0; i < k; i++)
    for (int j = 0; j < c[i]; j++) {
      a[pos] = i;
      pos++;
    }
}
```

$$n+n+k+n = 3n+k \sim O(n+k)$$

SHELL SORT

- Sort pairs of far away elems
- Sort the partially sorted array through insertion sort

1	3	9	6	2
1	3	9	6	2
1	2	9	6	3
1	3	9	6	2

```java
public static void shellSort(int[] a) {
  int n = a.length;
  int h = 1;
  while (h < n/3) h = 3 * h + 1;
  while (h >= 1) {
    for (int i = h; i < n; i++)
      for (int j = i; j >= h && a[j] < a[j - h];
             j = j - h) swap(a, j, j - h);
    h /= 3;
  }
}
```

BINARY SEARCH

TODO:
iterative vs recursive (Woche04 Aufgabe3)

```java
public static <T extends Comparable<T>> boolean bs(
    List<T> data, T target, int low, int high) {
  if (low > high) return false;
  int pivot = low + ((high - low) / 2);
  if (target.equals(data.get(pivot)))
    return true;
  else if (target.compareTo(data.get(pivot)) < 0)
    return searchBinary(data, target, low,pivot - 1);
  else
    return searchBinary(data, target, pivot+1, high);
}
```

$$O(\log n)$$

RECURSION

When a function calls itself. FP <3

LINEAR RECURSION

Recursive call starts ***at most one*** further recursive call

RECURSIVE → ITERATIVE

```java
static int[] reverseArrRec(int[] a, int i, int j) {
  if (i < j) {
    int temp = a[j];
    a[j] = a[i];
    a[i] = temp;
    reverseArray(a, i + 1, j - 1);
  }
  return a;
}
static int[] reverseArrIter(int[] a, int i, int j) {
  while (i < j) {
    int temp = a[j];
    a[j] = a[i];
    a[i] = temp;
    i = i + 1;
    j = j - 1;
  }
  return a;
}
```

TAIL RECURSION

Linear recursion where the recursive call is ***last***

```java
static int recsum(int x) {
  if (x == 0) return 0;
  else return x + recsum(x - 1);
  // no tail recursion because "+" is evaluated last
}
static int tailrecsum(int x, int total) {
  if (x == 0) return total;
  else return tailrecsum(x - 1, total + x);
}
```

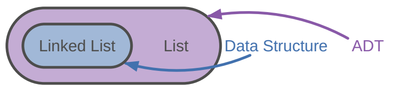
BINARY RECURSION

All non-terminal calls have ***two*** recursive calls

```java
static int binsum(int low, int high) {
  if (low > high)
    return 0;
  else if (low == high)
    return low;
  else {
    int mid = (low + high) / 2;
    return binsum(low, mid) + binsum(mid + 1, high);
  }
}
```

DATA STRUCTURES

ADT: Abstract Data Type (e.g. Interface)
- Describes "What", not "How"
- Describes Attributes, Operations and Exceptions
- Doesn't provide implementation details, that is done by the concrete data type



Data structure	*Access*	*Search*	*Insertion*	*Deletion*
Array	$O(1)$	$O(n)$	$O(n)$	$O(n)$
Linked List	$O(n)$	$O(n)$	$O(n)$	$O(n)$
Doubly Linked List	$O(n)$	$O(n)$	$O(1)$	$O(1)$
Stack	$O(n)$	$O(n)$	$O(1)$	$O(1)$
Queue	$O(n)$	$O(n)$	$O(1)$	$O(1)$
Binary Tree	$O(n)$	$O(n)$	$O(n)$	$O(n)$
Binary Search Tree	$O(n)$	$O(n)$	$O(n)$	$O(n)$
Hash Table	$O(n)$	$O(n)$	$O(n)$	$O(n)$

ARRAY

1	2	3	4

```java
int[] aAaAhrray = {69, 420, 1337};
```

LINKED LIST

1 → 2 → 3 → 4

```java
class LinkedList<T> {
  Node head;
  static class Node {
    T data;
    Node next;
    Node(T d) { data = d; }
  }
  public T insert(T data) {
    Node n = new Node(data);
    if (list.head == null)
      list.head = n;
    else {
      Node last = list.head;
      while (last.next != null)
        last = last.next;
      last.next = n;
    }
    return data;
  }
}
```

DOUBLY LINKED LIST

1 ⇄ 2 ⇄ 3 ⇄ 4

```java
class DoublyLinkedList<T> {
  Node head;
  static class Node {
    T data;
    Node next, prev;
    Node(T d) { data = d; }
  }
  public T insert(T data) {
    Node n = new Node(data);
    if (list.head == null)
      list.head = n;
    else {
      Node last = list.head;
      while (last.next != null)
        last = last.next;
      n.prev = last;
      last.next = n;
    }
    return data;
  }
}
```

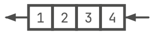
STACK

LIFO


```
class Stack<T> {
  private int maxSize;
  private T[] arr;
  private int top;
  public Stack(int size) {
    maxSize = size;
    arr = (T[]) new Object[maxSize];
    top = -1;
  }
  public void push(T value) {
    if (top == maxSize - 1)
      throw new StackOverflowError();
    arr[++top] = value;
  }
  public T pop() {
    if (top == -1)
      throw new StackUnderflowError();
    return arr[top--];
  }
}
```

QUEUE

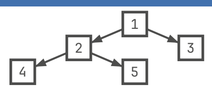
FIFO




```
class Queue<T> {
  private int maxSize;
  private T[] arr;
  private int front, rear;
  public Queue(int size) {
    maxSize = size;
    arr = (T[]) new Object[maxSize];
    front = rear = -1;
  }
  public T front() {
    if (front == -1)
      throw new StackUnderflowError();
    return arr[front];
  }
  public T rear() {
    if (rear == -1)
      throw new StackUnderflowError();
    return arr[rear];
  }
  public void enqueue(T value) {
    if (this.empty()) {
      front = 0;
      rear = 0;
      arr[front] = value;
    } else {
      front++;
      if (front < maxSize) arr[front] = value;
      else // rebuild array
    }
  }
  public T dequeue() {
    if (this.empty())
      throw new StackUnderflowError();
    T ret = arr[rear];
    if (front == rear) front = rear = -1;
    else rear++;
    return ret;
  }
  boolean empty() {
    return front == -1 && rear == -1;
  }
}
```

TREE

0 (Root)
1 (Inner)
2 (Leaf)

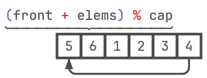



```
public class Tree<T> {
  private Node<T> root;
  public Tree(T rootData) {
    root = new Node<T>();
    root.data = rootData;
  }
  public static class Node<T> {
    private T data;
    private Node<T> left, right;
  }
}
```

REVERSING BINARY TREE

```
public static void swap(Node root) {
    if (root == null) return;
    Node tmp = root.left;
    root.left = root.right;
    root.right = tmp;
}
public static void mirror(Node root) {
    if (root == null) return;
    mirror(root.left);
    mirror(root.right);
    swap(root);
}
```

RING BUFFER




```
class RingBuffer {
  private int[] buffer;
  private int size;
  private int start;
  private int end;
  private boolean isFull;

  public RingBuffer(int size) {
    if (size <= 0)
      throw new IllegalArgumentException("size > 0");
    this.size = size;
    this.buffer = new int[size];
    this.start = 0;
    this.end = 0;
    this.isFull = false;
  }

  public void append(int item) {
    buffer[end] = item;
    end = (end + 1) % size;
    if (isFull)
      start = (start + 1) % size;
    if (end == start)
      isFull = true;
  }

  public List<Integer> getData() {
    List<Integer> items = new ArrayList<>();
    if (!isFull) {
      for (int i = 0; i < end; i++)
        items.add(buffer[i]);
    } else {
      for (int i = start; i < size; i++)
        items.add(buffer[i]);
      for (int i = 0; i < end; i++)
        items.add(buffer[i]);
    }
    return items;
  }
}
```

PRIORITY QUEUE

- Saves priority with each element
- Returns elements with lowest/highest prio first

Method	*Unsorted List*	*Sorted List*
insert	$O(1)$	$O(n)$
min	$O(n)$	$O(1)$
removeMin	$O(n)$	$O(1)$

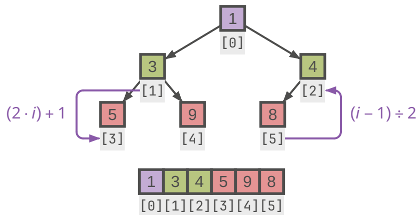
TODO:
impl

```
public interface Entry<K,V> {
  K getKey();
  V getValue();
}
public interface PriorityQueue<K,V> {
  int size();
  boolean isEmpty();
  Entry<K,V> insert(K key, V value);
  Entry<K,V> min();
  Entry<K,V> removeMin();
}
```

HEAP

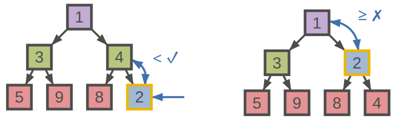
- Binary tree
- Layers $0, \ldots, h-1$ fully populated, layer h filled from left
- Operations always lowest RHS so that tree stays consistent

Min-Heap: Keys of nodes are **smaller** or equal than children's
Max-Heap: Keys of nodes are **larger** or equal than children's



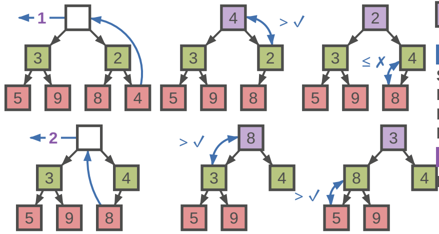
ENQUEUE

1) Insert into lowest RHS
2) Compare inserted elem recursively with parent
3) Swap if parent is larger, stop when parent is smaller



DEQUEUE

1) Remove root
2) Put lowest RHS into root pos
3) Compare moved elem recursively with smaller child
4) Swap if child is smaller, stop when no child is smaller
5) Tree is now heapified again, repeat at step 1



HEAP SORT

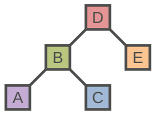
Dequeue elems from head into list, resorting tree on each iter

```
public class HeapSort {
  private static <T> int getLargest(T[] array, int n,
      int parent, Comparator<T> comparator) {
    int left = 2 * parent + 1;
    int right = 2 * parent + 2;
    int largest = parent;
    if (left < n && comparator.compare(array[left],
                              array[largest]) < 0)
      largest = left;
    if (right < n && comparator.compare(array[right],
                              array[largest]) < 0)
      largest = right;
    return largest;
  }

  static <T> void heapify(T[] arr,
      Comparator<T> comparator) {
    int n = arr.length;
    for (int i = n / 2 - 1; i >= 0; i--)
      heapify(arr, i, comparator);
  }

  public static <T> void heapify(T[] array,
      int parent, Comparator<T> comparator) {
    int largest = getLargest(array, array.length,
        parent, comparator);
    if (largest == parent) return;
    T temp = array[parent];
    array[parent] = array[largest];
    array[largest] = temp;
    heapify(array, largest, comparator);
  }

  static <T> void percolate(T[] array, int parent,
      int n, Comparator<T> comparator) {
    int largest = getLargest(array, n,
        parent, comparator);
    if (largest == parent) return;
    T temp = array[parent];
    array[parent] = array[largest];
    array[largest] = temp;
    percolate(array, largest, n, comparator);
  }

  static <T> void sort(T[] arr,
      Comparator<T> comparator) {
    int n = arr.length;
    heapify(arr, comparator.reversed());
    while (n > 1) {
      T temp = arr[0];
      arr[0] = arr[n - 1];
      arr[n - 1] = temp;
      n = n - 1;
      percolate(arr, 0, n, comparator.reversed());
    }
  }
}
```

BINARY SEARCH TREE (BST)

- All subnodes of **left** subtree are **smaller** than root
- All subnodes of **right** subtree are **larger** than root

TRAVERSING



Preorder (W-L-R): D B A C E
Postorder (L-R-W): A C B E D
Inorder (L-W-R): A B C D E
Breadth-First/ Level-Order: D B E A C

MORE DATA STRUCTURES

Set: Elements of same type, no duplicates, without order
Multiset: Set with duplicates
Map: KV-pairs with unique keys (put, get, remove $O(n)$)
Multimap: Key can have multiple values

HASHING

Hash-function h maps e onto $[0, n-1]$. $h(e)$ = position of e

$$e \xrightarrow{\text{hash code}} \mathbb{Z} \xrightarrow{\text{compression function}} \overbrace{[0, n-1]}^{\text{Index}}$$

Properties of good hash functions:
- Equal distribution
- Fast computation
- Constant time complexity
- Deterministic
- Incorporates as much information from key as possible

HASH FUNCTIONS

- Modulo primes $x \bmod p$
- Memory address (`Object.hashCode()` default)
- Byte/Integer cast `ByteBuffer.wrap(b).getInt()`
- Component sum (eg. sum string char codepoints)
- Polynomial accumulation $a_0 + a_1 z + a_2 z^2 + \ldots + a_{n-1} z^{n-1}$
  - $z = 33$ for strings
- Horner-Schema (identical, but easier to compute):
  - $a_0 + z \cdot (a_1 + z \cdot (a_2 + \ldots + z \cdot a_{n-1}))$

EQUALITY

It is generally necessary to override `hashCode` whenever `equals` is overridden

`x.equals(y) => x.hashCode() == y.hashCode()`

HASHING OBJECTS

- primitive types → builtin hash-code
- arrays → apply on each elem
- objects → apply recursively

```
// Horner schema with z = 31 and start = 17
@Override
public int hashCode() {
  int result = 17;
  for (Object field : fields) result = 31 * result
      + (field == null ? 0 : field.hashCode());
  return result
}
```

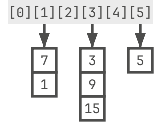
ANOMALIES

- Empty spaces
- Collisions
- Overflow

MANAGING HASH COLLISIONS

CLOSED ADDRESSING/SEPARATE CHAINING

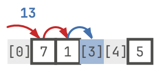
Each bucket is a list.
Search: $O(\text{entries}/\text{buckets}) = O(n/N)$
"Lastfaktor" $a = n/N$ (a can be > 1)
Large N: more storage, faster
Small N: less storage, slower



- Acceptable perf. under high load
- Larger storage overhead

OPEN ADDRESSING

e "overflows" into next empty bucket
→ Must probe 2 times in example



Problem: Primary Clustering

	put 89	put 69	put 10	put 58	put 11	put 12
0		69	69	69	69	69
1			10	10	10	10
2					11	11
3						12
4						
5						
6						
7						
8				58	58	58
9	89	89	89	89	89	89

Linear probing: $s(k, i) = h(k) + i$
Linear probing backwards: $s(k, i) = h(k) - i$
Quadratic probing: $s(k, i) = h(k) + i^2$

Deletion
- Only mark as "deleted"
- Move next best candidate to deleted position

Access

Probability for finding free spot:

n = entries $\qquad N$ = buckets

$a = \frac{n}{N}$ = probability: occupied $\qquad p$ = probing steps

$$P(X = p) = a^{p-1}(1 - a)$$

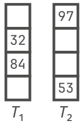
$$E(X) = \sum_{x=1}^{\infty} p \cdot P(X = p) = \frac{1}{1 - a} = \text{Nr. accesses}$$

a can never be bigger than 1

- No storage overhead
- Resize/Rehash compute intensive
- Bad performance under high load

CUCKOO-HASHING

$h_1(x) = x \bmod 5$ $\quad h_2(x) = x/5 \bmod 5$



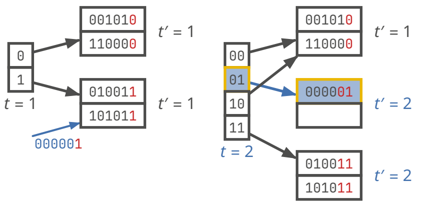
Insert: If $h_1(x) = \text{null} \Rightarrow T_1$, else push current y into T_2. If no cycle $\to O(1)$, if rehash is needed $\to O(n)$ (MAX_CYCLE)
Search: $O(1) \Rightarrow$ only has to check 2 positions
Rehashing: $x \bmod n \Rightarrow x \bmod (n+m)$

- constant time (if no rehashing) due to only 2 possible places
- no long probing chains
- potential of rehashing

EXTENDIBLE HASHING

- Resize/Rehash easier
- Good performance under high load
- Larger storage overhead

```
// getting index
int idx = key.hashCode() & ((1 << globalDepth) - 1);
// when to rehash
if (bucket.getLocalDepth() == globalDepth)
  growHashTable();
else
  splitBlock();
// splitting block
int highBit = 1 << localDepth;
for (Entry e : page.entries) {
    int h = e.key.hashCode();
    Page newPage = (h & highBit) != 0 ? p1 : p0;
    newPage.put(e.key, e.value);
}
```



DESIGN PATTERNS

Splitting algorithms and datastructures. Every time an OOP programmer has an idea of how to do something better they inevitably re-invent a concept from the functional paradigm.

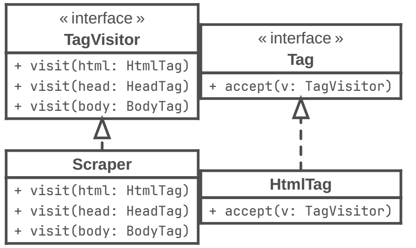
TYPES

Creational: Abstraction and instantiation (e.g., Factory, Singleton)
Structural: Composition of classes and objects into larger structures (e.g., Adapter, Facade)
Behavioral: Algorithms and distribution of responsibility among objects (e.g., Iterator, Visitor)

ITERATOR

```
interface Iterable<T> {
  Iterator<T> iterator();
}
public interface Iterator<E> {
  boolean hasNext();
  E next() throws NoSuchElementException;
  void remove() throws UnsupportedOperationException,
    IllegalStateException;
}

for (String s : stringList) { }        // ==
for (Iterator<String> i = stringList.iterator();
    i.hasNext()) String s = i.next(); // ==
Iterator<String> iter = stringList.iterator();
while (i.hasNext()) String s = i.next();
// Tree --> Depth-first Iterator
//      --> Breadth-first Iterator
```

VISITOR

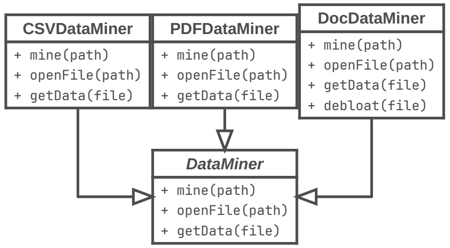


- separate algorithms and data
- keep algorithms in one place and not distributed over objs

```
public interface TagVisitor {
  void visit(HtmlTag html);
  void visit(HeadTag head);
  void visit(BodyTag body);
}
public class HtmlTag implements Tag {
  @Override
  public void accept(TagVisitor visitor) {
    visitor.visit(this);
    headTag.accept(visitor);
    bodyTag.accept(visitor);
    visitor.leave(this);
  }
}
```

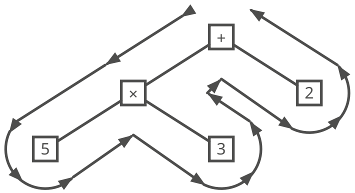
TEMPLATE METHOD

Break down an algorithm into a series of steps, turn these steps into methods, and put a series of calls to these methods inside a single template method.

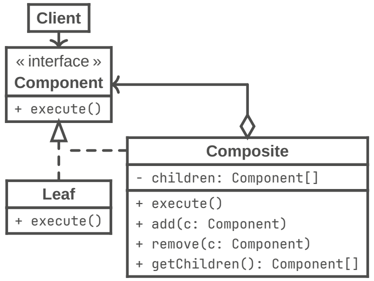


EULER TOUR TRAVERSING

Each node is traversed 3 times
- left (preorder)
- bottom (inorder)
- right (postorder)



COMPOSITE




```
interface Component {
  void printInfo();
}
class Item implements Component {
  private String name;
  public Item(String name) {
    this.name = name;
  }
  @Override
  public void printInfo() {
    System.out.println("Item: " + name);
  }
}
class Box implements Component {
  private List<Component> contents =
    new ArrayList<>();
  public void add(Component component) {
    contents.add(component);
  }
  public void remove(Component component) {
    contents.remove(component);
  }
  @Override
  public void printInfo() {
    for (Component c : contents)
      c.printInfo(); // Delegate to children
  }
}
```

ITERATORS

```
Iterator<String> it = stringList.iterator();
while (it.hasNext()) {
  String s = it.next();
  System.out.println(s);
}
```

Mutating Collection while iterating over it: ConcurrentModificationException

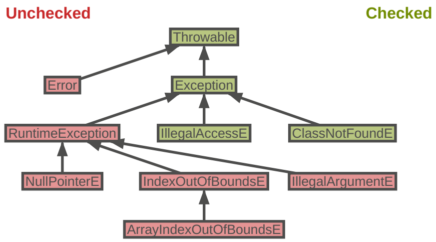
EXCEPTIONS

Error	*Exception*
Critical, don't handle	Runtime, handleable
OutOfMemoryError, StackOverflowError, AssertionError	IOException

Checked	*Unchecked*
Must be handled (or throws-declaration)	Not necessary
Checked by compiler	Compiler doesn't check
Exception, not RuntimeException	RuntimeException, Error

Child Exception gets caught in catch clause with parent class

```
void test() throws ExceptionA, ExceptionB {
  String c = clip("asdf");
  throw new ExceptionB("wack");
}

// finally ALWAYS executes, even on unhandled Exc.
try {
  test();
} catch (ExceptionA | ExceptionB e) {
  // ...
} finally { }

try { ... } catch(NullPointerException e) {
  throw e; // -->leaves blocks-->
} catch (Exception e) {
  // above e won't get caught!
} finally {
  // will still get executed
}

1 / 0; // ArithmeticException div by zero

String s = "";
s = null;
s.toUpperCase(); // NullPointerException

int[] arr = new int[] {1, 2, 3};
int elem = arr[8]; // ArrayIndexOutOfBoundsException
```



IMPORTANT STUFF

- Hashing should be added to equals fn's for strict equality
- Check if `input == null`
- Check if `array.length == 0`
- `IllegalArgumentException("reason")`
- try/catch finally block **always** executes

IO

```
try (var fr = new FileReader("text.txt")) {
  int input = fr.read();
  while (input >= 0) {
    if (input == ';') { /* do something */ }
    input = fr.read();
  }
}

try (FileWriter writer = new FileWriter("out.txt",
    StandardCharsets.UTF_8, true)) { // append
  writer.write("weeoo\n");
}

try {
  var input = new FileInputStream("text.txt");
  int i = input.read();
  while(i != -1) {
    System.out.print((char)i);
    i = input.read();
  }
  input.close();
} catch (Exception e) {
  e.printStackTrace();
}

try (BufferedReader reader = new BufferedReader(
    new FileReader("text.txt",
      StandardCharsets.UTF_8))) {
  String line;
  while ((line = reader.readLine()) != null) {
      System.out.println(line);
  }
}

try (
    FileReader reader = new FileReader("in.txt");
    FileWriter writer = new FileWriter("out.txt")
  ) {
  int i = input.read();
  while(i >= 0) {
    writer.write(i);
    i = input.read();
  }
}
```

TRY WITH

```
try (var output = new FileOutputStream("f.txt")) {
  output.write("Hello".getBytes());
} catch (IOException e) {
  System.out.println("Error writing file");
} finally {
  System.out.println("Done");
}
```

SERIALIZING

```
class X implements Serializable { }
// Serializing
try (var stream = new ObjectOutputStream(
    new FileOutputStream("s.bin"))) {
  stream.writeObject(new X());
}
// Deserializing
try (var stream = new ObjectInputStream(
    new FileInputStream("s.bin"))) {
  X x = (X) stream.readObject();
}
```

FUNCTION

```
public interface Function<T, R> {
  R apply(T t);

  static <T> Function<T, T> identity();

  <V> Function<T, V> andThen(
    Function<? super R, ? extends V> after);

  <V> Function<V, R> compose(
    Function<? super V, ? extends T> before);
}
```

PREDICATE

```
public interface Predicate<T> {
  boolean test(T t);
}

static void removeAll(Collection<Person> collection,
    Predicate criterion) {
  var it = collection.iterator();
  while (it.hasNext())
    if (criterion.test(it.next()))
      it.remove();
}
```

COMPARABLE

```
public interface Comparable<T> {
  int compareTo(T obj);
}

var l = new ArrayList<Integer>(asList(3,2,4,5,1));
l.sort((a, b) -> a > b ? 1 : -1); // ==
l.sort((a, b) -> a - b);          // 1,2,3,4,5

class Person implements Comparable<Person> {
  private final String firstName, lastName;
  @Override
  public int compareTo(Person other) {
    int result = lastName.compareTo(other.lastName);
    if (result == 0)
      result = firstName.compareTo(other.firstName);
    return result;
  }

  static int compareByAge(Person a, Person b) {
    return Integer.compare(a.getAge(), b.getAge());
  }
}
List<Person> people = ...;
Collections.sort(people);
people.sort(Person::compareByAge);
```

COMPARATOR

```
class AgeComparator implements Comparator<Person> {
   @Override
   public int compare(Person a, Person b) {
     return Integer.compare(a.getAge(), b.getAge());
   }
}
Collections.sort(people, new AgeComparator());
people.sort(new AgeComparator());

people.sort(Comparator
  .comparing(Person::getAge)
  .thenComparing(Person::getFirstName)
  .reversed())

Comparator.comparing(Person::getName,
  (s1, s2) -> s2.compareTo(s1));
// ==
Comparator.comparing(Person::getName).reversed();

Comparator<T> nullsLast(Comparator<T> c);
Comparator<T> nullsFirst(Comparator<T> c);
Comparator<T> comparing(Function<T,U> keyExtractor,
  Comparator<U> c);
Comparator<T> comparingInt(ToIntFunction<T,U> f);
```

FUNCTIONALINTERFACE

Any interface with a single abstract method is a functional interface

```
@FunctionalInterface
public interface ShortToByteFunction {
    byte applyAsByte(short s);
}
@FunctionalInterface
public interface PersonStringifier {
    String getNameAndAge(Person p);
}
```

COLLECTION

```
boolean add(E e);
boolean remove(Object o);
boolean equals(Object o);
int hashCode();
int size();
boolean isEmpty();
Object[] toArray();
void clear();
boolean contains(Object o);
boolean addAll(Collection<? extends E> c);
boolean containsAll(Collection<?> c);
boolean removeAll(Collection<?> c);
boolean retainAll(Collection<?> c);

Set<String> noDup = new HashSet<>();
```

COLLECTION IMPLEMENTATIONS

```
// List
int indexOf(Object o);
int lastIndexOf(Object o);
E get(int index);
subList(int from, int to);
void sort(Comparator<? super E> c);

// Stack
E peek();
E pop();
E push(E item);
boolean empty();
int search(Object o);

// Queue
E element();      // throws -> peek();     doesn't
E remove();       // throws -> poll();     doesn't
boolean add(E e); // throws -> offer(E e); doesn't

// Set
// (I) SortedSet -> (C) TreeSet
// (C) HashSet, (C) LinkedHashSet

// Map
// HashMap
boolean containsKey(Object key);
boolean containsValue(Object value);
Set<Map.Entry<K, V>>> entrySet();
V get(Object key);
V put(K key, V value);
V putIfAbsent(K key, V value);
V replace(K key, V value);
V remove(Object key);
V getOrDefault(Object key, V defaultValue);
Set<K> keySet();
Collection<V> values();
```

LAMBDAS

```
String pattern = readFromConsole();
//     vvv not final -> Error
while (pattern.length() == 0)
  pattern = readFromConsole();
Utils.removeAll(people, p ->
    p.getLastName().contains(pattern));
// local variable ... referenced from a lambda
expression must be final or effectively final

// Predicate      :: a -> boolean
Predicate<Integer> isLarge = (v) -> v > 69420;
// Function       :: a -> b
Function<Integer, String> str = (v) -> "" + v;
// Supplier       :: a
Supplier<String> hello = () -> "Hello, World!";
// Consumer       :: a -> void
Consumer<Integer> consoomer = (v) -> log(v);
// UnaryOperator  :: a -> a
UnaryOperator<Integer> more = (v) -> v * v;
// BinaryOperator :: a -> a -> a
BinaryOperator<Integer> less = (a, b) -> a - b;
```

STREAMS

```
import java.util.stream.*;

people
  .stream()
  .distinct()
  .filter(p -> p.getAge() >= 18)
  .skip(5)
  .limit(10)
  .map(p -> p.getLastName())
  .sorted()
  .forEach(System.out::println);

people
  .stream()
  .reduce(0, (acc,cur) -> acc + cur.getAge());

list.stream().mapToInt(Integer::intValue);
list.stream().mapToInt(Integer::parseInt);
```

OPTIONAL (HASKELL / RUST IN BLOAT)

```
T get(); // NoSuchElementException
boolean isPresent();
void ifPresent(Consumer<T> consumer);
T orElse(T other);
static Optional<T> empty();
static Optional<T> of(T value);
```

METHODS

```
boolean allMatch(Predicate<T> predicate);
boolean anyMatch(Predicate<T> predicate);
boolean noneMatch(Predicate<T> predicate);
static Stream<T> concat(Stream<T> a, Stream<T> b);
Stream<T> distinct();
Stream<R> flatMap(Function<T, R> mapper);
Stream<T> limit(long maxSize);
Stream<T> skip(long maxSize);
Stream<sorted>(Comparator<T> comparator);
```

TERMINAL OPERATIONS

```
Optional<T> min(Comparator<T> comparator);
Optional<T> max(Comparator<T> comparator);
Optional<T> findAny();
Optional<T> findFirst();
void forEach(Consumer<T> action);
long count();
void forEachOrdered(Consumer<T> action);
// IntStream
average();
sum();
```

COLLECTORS

```
// List
s.collect(Collectors.toList());
// TreeSet
s.collect(Collectors.toCollection(TreeSet::new));
// String
s.collect(Collectors.joining(", "));
// Integer
s.collect(Collectors.summingInt(Person::getAge));
// Map<String, Person>
s.collect(Collectors.groupingBy(Person::getCity));
// Map<String, Integer>
s.collect(Collectors.groupingBy(Person::getCity,
  Collectors.summingInt(Person::getSalary)));
// Map<boolean, List<Person>>
s.collect(Collectors.partitioningBy(s ->
  s.getAge() > 18))
```

MORE API'S WE “SHOULD” BE PROVIDED WITH

Yes I'm salty, thanks for pointing that out

```
// Integer
static int parseInt(String s);
// System
arraycopy(Object src, int srcPos, Object dest,
    int destPos, int length);
```